HEART DISEASE PREDICTION WITH SHAP AND EXPLAINABLE APPROACH

A Thesis

Submitted in partial fulfillment of the requirements for the award of the Degree of

Masters of Technology

IN

Computer Science and Engineering

BY

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This is to certify that the work embodied in this thesis entitled **“**Heart disease prediction with SHAP and Explainable approach**”**, is carried out by **Kumari Anjali Suman (MT/CS/10016/23)** has been approved for the degree of MASTERS OF TECHNOLOGY (Computer Science and Engineering) of Birla Institute of Technology, Mesra, Ranchi.

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ABSTRACT

This study explores the integration of SHAP (SHapley Additive exPlanations) with random forest models to interpret predictions in cancer detection. SHAP offers both global insights into feature importance and local explanations for individual predictions, addressing the need for transparency in complex models. By analyzing feature contributions and their interactions, the research highlights how SHAP enables clinicians to understand model decisions better, fostering trust and aiding in decision-making. The approach demonstrates the potential for interpretable machine learning in advancing cancer diagnostics. The research highlights SHAP's capability to address the "black-box" nature of random forests by explaining individual predictions and the overall impact of features on the model. This dual interpretability facilitates a better understanding of complex relationships between features and outcomes, contributing to advancements in precision medicine and personalized treatment strategies. Integrating SHAP with cancer detection models underscores the importance of transparent and reliable machine learning tools in clinical applications

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Table of Contents

Chapter-1

INTRODUCTION

1.1.Background and the need for predictive model

Heart disease remains a global health challenge, being one of the leading causes of death and a significant burden on healthcare systems. Traditional diagnostic approaches rely on medical tests and clinical expertise but often face limitations in speed and accuracy. Machine learning models like Random Forest (RF) have emerged as transformative tools for improving diagnostic efficiency by analyzing complex datasets and uncovering patterns in patient health indicators, such as cholesterol, blood pressure, and other vital factors.

However, the adoption of machine learning in healthcare also raises concerns about model transparency and trust. To address these, SHAP (SHapley Additive exPlanations) provides a framework for interpreting machine learning models, offering both global and local insights into predictions. By explaining how specific features contribute to an RF model's decision, SHAP bridges the gap between advanced computational techniques and their practical use in clinical environments. This study highlights the integration of RF and SHAP, emphasizing their role in enhancing heart disease detection while ensuring interpretability and trustworthiness.

Detecting heart disease presents several challenges due to the complexity and variability of the condition.

- **Symptom Variability**: Heart disease often manifests through diverse symptoms, which can be mild or mimic other illnesses, complicating early detection.

- **Data Complexity**: Accurate diagnosis requires interpreting vast and varied health data, including imaging, lab results, and clinical history.
- **Late Diagnosis**: Patients often seek medical attention only after significant disease progression.
- **Health Disparities**: Access to advanced diagnostic tools varies globally, limiting early detection in underserved areas.
- **Comorbid Conditions**: Overlapping symptoms with other illnesses can obscure diagnosis.

<u>The need for predictive models</u>

Heart disease detection relies heavily on predictive models to address limitations in traditional diagnostic methods. These models utilize machine learning and statistical techniques to analyze complex datasets, uncovering patterns and risk factors that might be missed through conventional means. Predictive tools enable earlier detection, allowing for timely medical intervention and personalized treatment plans. They also reduce the dependency on invasive and expensive diagnostic procedures, increasing efficiency in healthcare systems. By improving accuracy and speed, predictive models significantly enhance patient outcomes and support preventive healthcare strategies.

In the medical field, predictive accuracy alone is insufficient. It is equally important to ensure that models are interpretable, so healthcare professionals can trust and understand the predictions. Tools like SHAP (SHapley Additive exPlanations) offer insights into how individual features contribute to model predictions, addressing the "black-box" nature of many ML models.

The integration of predictive models like Random Forest with interpretability tools such as SHAP holds great promise in transforming heart disease detection and improving patient outcomes.

1.2. Overview of ML in healthcare

Machine learning (ML) in healthcare is revolutionizing the industry by enabling data-driven insights, personalized treatments, and efficient diagnostic processes. ML algorithms analyze vast amounts of healthcare data, including medical records, imaging, and genetic information, to predict outcomes and suggest interventions. Applications include disease diagnosis, treatment recommendation, and patient monitoring, offering faster and more accurate solutions compared to traditional methods.

Predictive models powered by ML are particularly impactful in early disease detection, such as identifying cancer or predicting heart conditions. Algorithms like neural networks and decision trees assist in analyzing complex medical data, improving diagnostic precision. Natural language processing (NLP) further enhances healthcare by extracting valuable insights from unstructured text, such as clinical notes and research articles.

Another significant contribution is in drug discovery and development. ML accelerates the identification of potential drug candidates by simulating biological interactions, significantly reducing time and costs. Additionally, wearable devices and IoT sensors integrate ML to monitor patient health in real-time, supporting preventive care and reducing hospital visits.

Despite its transformative potential, challenges remain, including data privacy concerns, bias in algorithms, and the need for standardized integration into clinical workflows. However, ongoing advancements in explainable AI and regulatory frameworks aim to address these hurdles, ensuring the safe and ethical adoption of machine learning in healthcare.

1.3.Random Forest (RF) and Its Advantages in Classification Tasks

Random Forest (RF) is a popular ensemble learning algorithm designed to enhance the predictive power of individual decision trees by combining their outputs. Developed as an extension of the decision tree approach, RF generates multiple trees using subsets of the data and features, a process known as bagging (bootstrap aggregation). Each tree provides a prediction, and the final output is determined by averaging (in regression) or majority voting (in classification). This ensemble approach minimizes overfitting and ensures more reliable, generalizable predictions.

RF's ability to handle high-dimensional datasets with a mix of categorical and numerical variables makes it particularly suitable for classification tasks. For example, in healthcare, RF can classify diseases based on patient symptoms and lab results, while in finance, it predicts creditworthiness or fraud detection. The algorithm is also robust to missing or noisy data, as each tree in the forest relies on a random sample of features, reducing the impact of anomalies.

One of the standout features of RF is its capacity to measure feature importance, offering insights into which variables most influence predictions. This characteristic is crucial for interpretability, especially in applications where understanding the model's reasoning is essential, such as regulatory environments or medical diagnostics. Moreover, RF is computationally efficient and scalable, capable of handling large datasets due to its parallelizable nature, which allows each tree to be trained independently.

Despite its strengths, RF is not without limitations. It can sometimes be less interpretable than simpler models due to the complexity of combining multiple trees. Additionally, when applied to datasets with a large number of irrelevant features, the algorithm's performance can be hindered. Nonetheless, RF remains a powerful, versatile, and widely used tool for classification tasks, providing a balance of accuracy, robustness, and ease of use.

1.4. Objective

The study titled "Explaining Random Forest Heart Disease Detection Prediction with SHAP: A Study of Global and Local Interpretation" aims to demonstrate how interpretability enhances the utility of machine learning in healthcare. It investigates the effectiveness of the Random Forest algorithm for detecting heart disease while leveraging SHAP (SHapley Additive exPlanations) to explain the model's decisions. The research compares global interpretability, which highlights overall feature importance, with local interpretability, focusing on individual predictions, providing actionable insights for personalized treatment plans.

This study emphasizes bridging the gap between machine learning and healthcare professionals by promoting trust and transparency. By examining feature contributions, it enables clinicians to understand which attributes, such as cholesterol levels or blood pressure, significantly impact predictions. This approach not only supports diagnostic decisions but also aligns machine learning tools with ethical considerations and patient outcomes, ultimately improving the adoption of AI in sensitive domains like heart disease detection.

Chapter-2

LITERATURE REVIEW

A literature review examines existing research to understand the development, effectiveness, and application of classification algorithms in various domains. Key topics include the exploration of classical methods like decision trees, support vector machines (SVM), and k-nearest neighbors (KNN), as well as advanced techniques like ensemble learning (random forests and boosting) and neural networks. The review often discusses algorithmic performance, feature selection strategies, and evaluation metrics like accuracy, precision, recall, and F1 score.

The work also focuses on domain-specific applications, such as disease diagnosis, fraud detection, and image classification, highlighting challenges like handling imbalanced datasets, overfitting, and computational complexity. Recent trends emphasize hybrid models and explainability tools, like SHAP and LIME, which enhance trust and interpretability in classifiers, especially in critical fields like healthcare and finance.

2.1 Related work based on machine learning classifier

Tsehay [1] used KNN technique on breast cancer dataset consists of 1025 observations with 13 features. The challenges of biased classification on imbalanced observation are solved using decision tree and adaptive boosting algorithms. The comparison results shows that adaptive boosting technique has better performance than decision tree.

S B Imandoust et al. [3] reviewed the comprehensive study related to K Nearest Neighbor with their historical & theorical background, benefits & challenges.

Manav Sharma et.al.[4] exploited Convolutional Neural Network (CNN) algorithm on 253 brain MRI images dataset. The accuracy of proposed model is 97.79% when applied on training dataset whereas 82.86% carried out on validation dataset. The authors also concluded that the losses gradually decrease in increase of number of epochs.

Grandhi and M. Adimoolam [5], a comparative analysis on predictive performance of machine learning techniques i.e., Artificial Neural Network (ANN) and Support Vector Machine (SVM). The classifiers are evaluated against their prediction accuracy and the results shows maximum accuracy of ANN with 97.2% whereas 93.2% is of SVM accuracy.

Md. Shoheb, Y. Sreenivasulu, et.al., [2], Urdu spam e-mail detection model is proposed by employing various machine learning and deep learning algorithms such as Support Vector Machine (SVM), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM) and Naïve Bayes on dataset form Kaggle containing 5000 emails. The author experimented two different observations and the results show Naïve Bayes is better in terms of recall & f-measures and Support Vector Machine (SVM) achieves better output in precision. The higher prediction accuracy is acquired by Long Short-Term Memory (LSTM) of 98.4% with less loss rate of 5%.

Gopal Deshmukh, Bipin Sule, et.al.[6], used 569 breast cancer cases from the Wisconsin Breast Cancer (Diagnostic) dataset with 569 instances and 30 attributes. The prediction model used three different machine learning classifiers; Random Forest, Linear Regression and decision tree. The comparative analysis shows higher accuracy of decision tree technique with 99% as training accuracy and 95% as testing accuracy.

For the prediction of breast cancer diagnostic, Autsuo Higa [7] proposed two powerful data mining technique, Decision tree and Neural network. The comparative study shows neural network model with higher rate of classification i.e., 95.9%.

Asif et.al (2021) proposed a machine learning-based model for detection and classifying the COVID19 cases by examining the chest x-ray scans. The dataset is taken from Kaggle containing 550 total scans and splitting 75% as training and 25% as testing set. Two well-known machine learning classifiers support vector machine and random forest were used in this dataset. Support vector machine show an accuracy of 99.27% and random forest show an accuracy of 96.89%. The author also calculated precision, recall and compared accuracy of proposed model with their literature review models.

2.2. Related work based on healthcare prediction

In the research work of KM Jyoti Rani (2020) [21] for diabetes prediction, K Nearest Neighbor, Logistic Regression, Decision Tree, Support Vector machine with several kernels is used and compared their accuracies of both training and testing set. Experimental results determine 99% of accuracy achieving in using decision tree algorithm.

Chintan et.al., (2023) [22] aims to improve the classification accuracy of different machine learning classifiers such as decision tree, random forest, XGBoost, multilayer perceptron. This research develops a model that can correctly predict cardiovascular diseases to reduce the fatality caused by it. Using the dataset consisted of 70,000 rows and 12 attributes and cross validation approach, the study results the highest accuracy rate of 87.28% by using multilayer perceptron and lowest accuracy of 86.37% given by decision tree.

Chandan et.al., (2021) [24] aims to discover a model for prediction of thyroid disease with higher accuracy. The dataset was taken from UCI (University of California Irvine machine learning repository) containing 215 samples and 5 features. K-Nearest Neighbor, support vector machine, artificial neural network, decision tree and logistic regression were taken and measured their accuracy.

Higher accuracy obtained from logistic regression model (96.92%) and further concluded that this model will be considered for their prediction model.

For the prediction of thyroid disease, Ankita et.al (2018) [23] chosen four machine learning classifiers such as decision tree, ANN (Artificial Neural Network), support vector machine, K-NN (Nearest Neighbor). The author examined their accuracy and mean squared error of each algorithm also.

Madhumita and Smita (2021) [26] experimented a data mining model for predicting heart disease. The data was taken from Kaggle site (303 rows and 14 columns). For implementation of dataset, python programming was used in Jupyter notebook. The author used random forest data mining algorithm was implementation and 10-fold cross validation technique for splitting data. In proposed work, 86.9% classification accuracy is obtained for prediction of heart disease with diagnosis rate 93.3%.

Raniya Rone Sarra (2023) [26] published a paper for predicting a heart disease by using ANN, SVM and DNN with 93.44%, 86.88% and 87% accuracy respectively using 303 samples.

Singh et al., (2021) [25] predicted breast cancer using K Nearest Neighbor. The dataset collected from WBC datasets (Wisconsin breast cancer) with 699 instances (Training Set with 468 data values and Testing Set with 231 data points). Dimension reduction techniques is also aimed to apply for classification models.

2.3. Related work based on explainable models (2020)

P. Linardatos, et.al., [8] conducted a literature survey on existing machine learning interpretability methods. The author focuses on four major categories; explaining black-box model, creating white-box model, uprightness and regulate discrimination and identifying sensitivity of existing model prediction.

For the systematic literature review, Saranya A. and Subhashini R. [9] followed 91 articles on development of XAI and different applications published between 2018-2022. Highlighting four common approaches used to build XAI taxonomy, the author explained enormously about explainable AI that embolden researchers to develop innovative methods.

A. Bozorgpanah & V. Torra [10] used three different multivariate data sets namely, Cervical cancer (risk factors), Breast cancer Coimbra and USA House and compared the model build from original and masked datasets. MDAV and noise addition are considered as two masking techniques. The comparative analysis shows MDAV method as best performer on the basis of three different metrices,i.e.,Irregularity, Utility, and Rank correlation.

Mehmed Kantardzic et.al.,[11] reviewed two different explainable methods (ad-hoc & post-hoc) for cancer detection using MRI images. Additionally, the author emphasized the importance of explainable technique with respect to clinical deem and a gap between medical methods requirement and provided.

Scott M. Lundberg and Su-In Lee [17] have been done a comprehensive survey and used different estimation methods on SHAP framework including kernel SHAP, Linear SHAP, Deep SHAP and many more to show that there is a unique solution that meets desirable principles in approach for model interpretation and can be helpful for developing new innovative methods in future.

Dr. I. B. G, Prof. G. M., et.al.,[18] accomplished extensive survey on two widely used explainable AI namely, SHAP and LIME, which describes model outcome prediction in both locally and globally. The author discussed and highlighted their important and critical points that should know before applying it on complex models.

D. Devegonda, V. Jayaram, et.al.,[19] used Gulf of Mexico dataset to study the importance of seismic attributes. A SHAP was implemented on Random Forest classifier to differentiate between salt and MTDs and to know the significant impact of four input candidate attributes i.e., total energy, coherence, GLCM entropy, and reflector convergence., in model’s prediction outcome. The impact on model’s output shows the low value of total energy and coherence whereas high values attributes shown by GLCM entropy and reflector.

2.4. SHAP (SHapley Additive exPlanations):An interpretability tool

SHAP (SHapley Additive exPlanations) is a robust interpretability tool rooted in game theory, designed to explain the predictions of machine learning models. It assigns an importance value to each feature by comparing a model's prediction with and without that feature. This approach ensures fairness and consistency in explanations. SHAP supports both global and local interpretability, providing insights into overall feature importance across a dataset and individual predictions. It is model-agnostic, making it applicable to various machine learning algorithms.

One of SHAP's key strengths is its adherence to theoretical properties, such as efficiency and symmetry, which make it unique among interpretability methods. By visualizing feature contributions through summary plots, force plots, and dependence plots, SHAP makes complex models more transparent. It is widely applied in fields like healthcare, finance, and scientific research, where understanding model decisions is critical. SHAP enhances trust and accountability in AI-driven systems, enabling informed decision-making and addressing ethical concerns.

2.5. Comparative study of global and local prediction Methods

In SHAP (SHapley Additive ExPlanations), global predictions involve understanding the overall impact of model by analyzing feature contributions across the entire dataset. This helps identify which features are most influential on average. In contrast, local predictions focus on explaining individual predictions, showing how each feature contributes to the specific outcome for a single instance. By combining global insights for model trends with local explanations for instance-specific details, SHAP provides a comprehensive interpretability framework for understanding model predictions.

Global Interpretability

Global interpretability focuses on understanding the overarching behaviour of a machine learning model. It explains how features influence predictions on average across an entire dataset. Tools like Feature Importance and Partial Dependence Plots (PDPs) are popular for global interpretability. These methods provide a high-level view of feature contribution but may fail to capture nuances of specific instances, making them more suited for gaining a broader understanding of the model.

Advantages:

- Offers insights into model-wide patterns and trends.
- Useful for debugging, validating models, and communicating general findings.

Challenges:

- Oversimplifies localized interactions.
- May not explain individual predictions effectively.

Local Interpretability

Local interpretability delves into the decision-making process of a model for specific instances. SHAP (SHapley Additive ExPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) are widely used methods. These approaches aim to provide explanations tailored to a single data point, clarifying the factors behind its prediction.

Advantages:

- Enables instance-specific analysis, aiding in critical decisions (e.g., healthcare or fraud detection).
- Enhances trust in the model by offering clear, instance-level reasoning.

Challenges:

- Computationally expensive, especially for complex models.
- Explanations may lack scalability to the entire dataset.

Chapter-3

METHODOLOGY

3.1 Data Collection and Preprocessing

The dataset was taken from Kaggle site https://www.kaggle.com/datasets/rashikrahmanpritom/heartattackanalysis-prediction-dataset?select=heart.csv containing 303 cases of heart attack patients with 14 feature attributes. For this experiment, selected algorithms code is done using python programming language in Anaconda Jupyter Notebook with default kernel Python 3(ipykernel), Google colab and many useful python libraries. The main objective is to analyze and predict that whether a patient will suffer from heart attack or not.

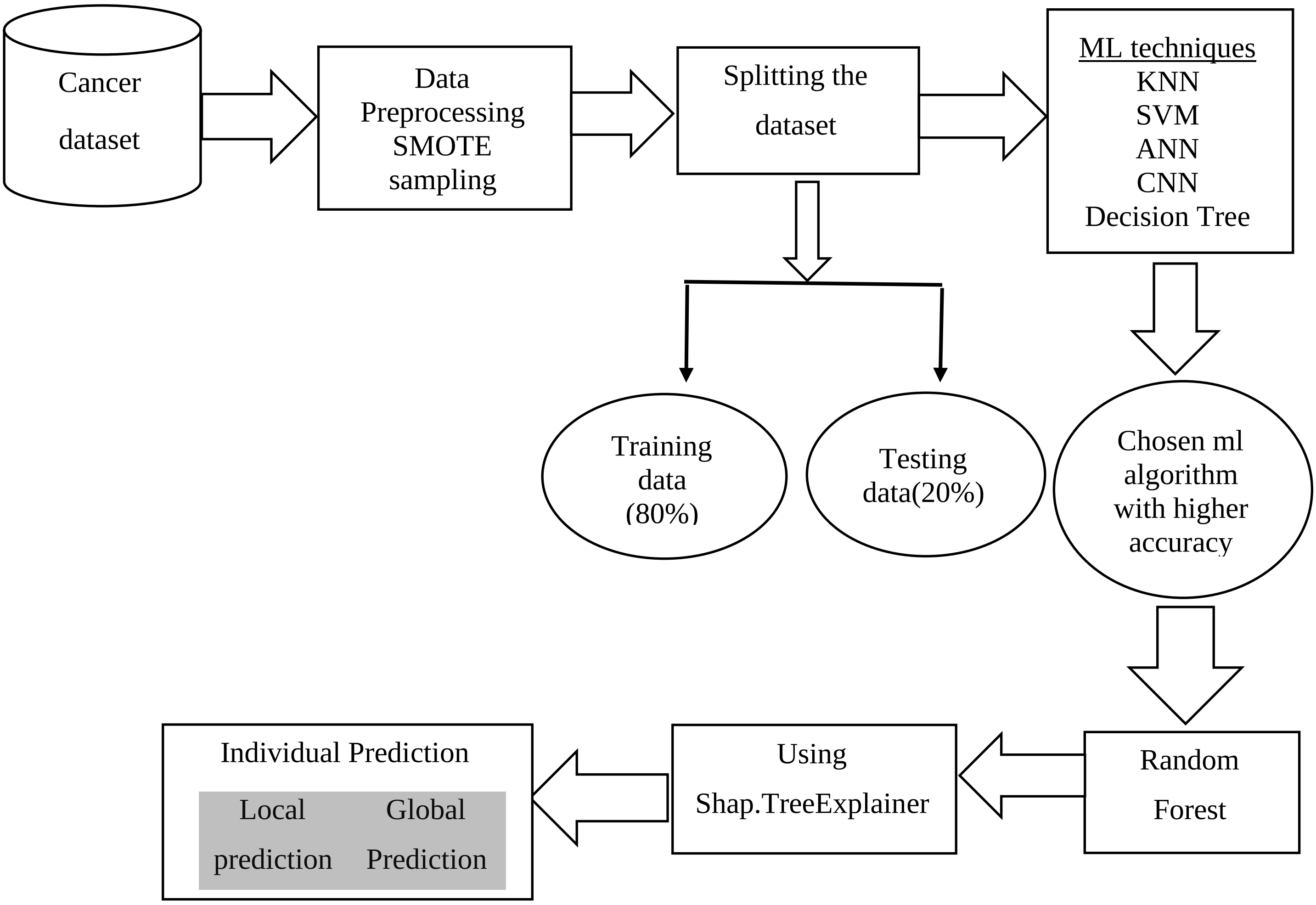
Features label and their description

Label	Description
age	Age of the patient
sex	Sex of the patient
cp	Chest Pain type chest pain type o Value 1: typical angina o Value 2: atypical angina o Value 3: non-anginal pain o Value 4: asymptomatic
trtbps	resting blood pressure (in mm Hg)
chol	cholesterol in mg/dl fetched via BMI sensor
fbs	(fasting blood sugar > 120 mg/dl) (1 = true; 0 = false)
rest-ecg	resting electrocardiographic results o Value 0: normal o Value 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV) o Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria
thalachh	maximum heart rate achieved

exng	exercise induced angina (1 = yes; 0 = no)
oldpeak	Previous peak
slp	the slope of the peak exercise ST segment
caa	number of major vessels (0-3)
thall	1= normal; 2 = fixed defect; 3 = reversable defect
output	0= less chance of heart attack 1= more chance of heart attack

After the collection of dataset, first part is to check garbage data. It may be possible that input can contain unnecessary or null data which becomes a reason to deteriorate an accuracy, thus it should be encountered in data cleaning to take off such unwanted data. In our chose dataset, no garbage data was presented so we didn’t follow this step. Cleaned data is used in both training and testing phase which are feds as input in the algorithm.

Training and testing data was splitted into 80% and 20% respectively but the problem raised i.e., imbalance of data which means the counts of ‘1’ is 133 whereas 109 is the counts of ‘0’. For balancing it, SMOTE (Synthetic Minority Oversampling Technique) method is used which given the result 133 as both counts of ‘0’ and ‘1’. Data preprocessing like standardization or normalization of numerical feature is also done in order to set them in common scale.



Some well-known machine learning and deep learning algorithms are implemented upon dataset to learn and classify/predict that whether a patient is having less chance of heart attack or more chance of heart attack.

For examine and compare the performance of predicted accuracy, five different machine learning algorithms are used, namely., Random Forest, Decision tree, ANN (Artificial Neural Network), CNN (Convolutional Neural Network) and SVM (Support Vector Machine). Random forest shown the highest accuracy, so SHAP XAI was implemented on that.

3.2. Explanation of SHAP with Random Forest

SHAP (SHapley Additive exPlanations) is a model-agnostic interpretability tool used to explain predictions of complex models like Random Forests. SHAP attributes importance to features by computing Shapley values based on game theory, reflecting each feature's contribution to the prediction. When combined with Random Forests, SHAP enables both global interpretability (analyzing overall feature importance) and local interpretability (explaining individual predictions). This dual approach helps in understanding how specific data points and general patterns contribute to model outputs.

Steps to Integrate SHAP with Random Forest

- **Train the Random Forest Model**

Build and train a Random Forest model using your dataset to predict the desired outcome.

- **Import SHAP Library**

Install and import the SHAP package alongside necessary libraries like scikit-learn.

- **Initialize the Explainer**

Use shap.TreeExplainer() to create an explainer object for the trained Random Forest model.

- **Compute SHAP Values**

Call explainer.shap_values() on the dataset to calculate SHAP values for each instance.

- **Visualize Results**

Use SHAP's plotting functions like summary_plot, force_plot, and dependence_plot to analyze and visualize the feature importance and individual predictions.

- **Integrate in Workflow**

This involves leveraging SHAP's outputs, such as feature importance rankings or explanations of individual predictions, to enhance decision-making or improve the model's utility.

Using SHAP with Random Forest models is done with tools like Python, the SHAP library, and Scikit-learn. Python helps in coding, while Scikit-learn builds the Random Forest model. The SHAP library explains the model by showing how each feature affects predictions. Visual tools like Matplotlib, seaborn make it easy to understand the results. To process larger datasets or complex tasks, good hardware like fast computers or cloud services ensures quick and smooth analysis. These tools and resources make the entire process efficient and insightful.

Chapter-4

EXPERIMENTAL RESULTS

In this section, the experimental test results on the proposed models are explained. The predictive performance of all five algorithms is analyzed by employing the performance metrics such as accuracy and confusion matrix.

4.1. Model Performance Evaluation

In initial evaluation, we performed the experiment and compared their performance metrices with balancing and without balancing for all models as chosen parameters strongly impact the durability and adaptability capability of machine learning algorithms. We used binary crossentropy as loss function in ANN to calculate four different parameters i.e., accuracy, precision, recall and f1 score.

Without use of SMOTE sampling

Algorithm	Accuracy	F1	Precision	Recall
KNN	80.32%	79.31%	88.46%	71.875%
SVM	77.04%	78.125%	78.125%	78.125%
DT	83.60%	82.75%	92.30%	75%
RF	85.24%	86.15%	84.44%	87.5%
ANN	83.60%	83.87%	86.66%	81.25%

With use of SMOTE method

Algorithm	Accuracy	F1	Precision	Recall
ANN	80.32%	79.31%	88.46%	71.87%
SVM	83.60%	84.37%	89.65%	81.25%
DT	80.32%	80%	85.71%	75%

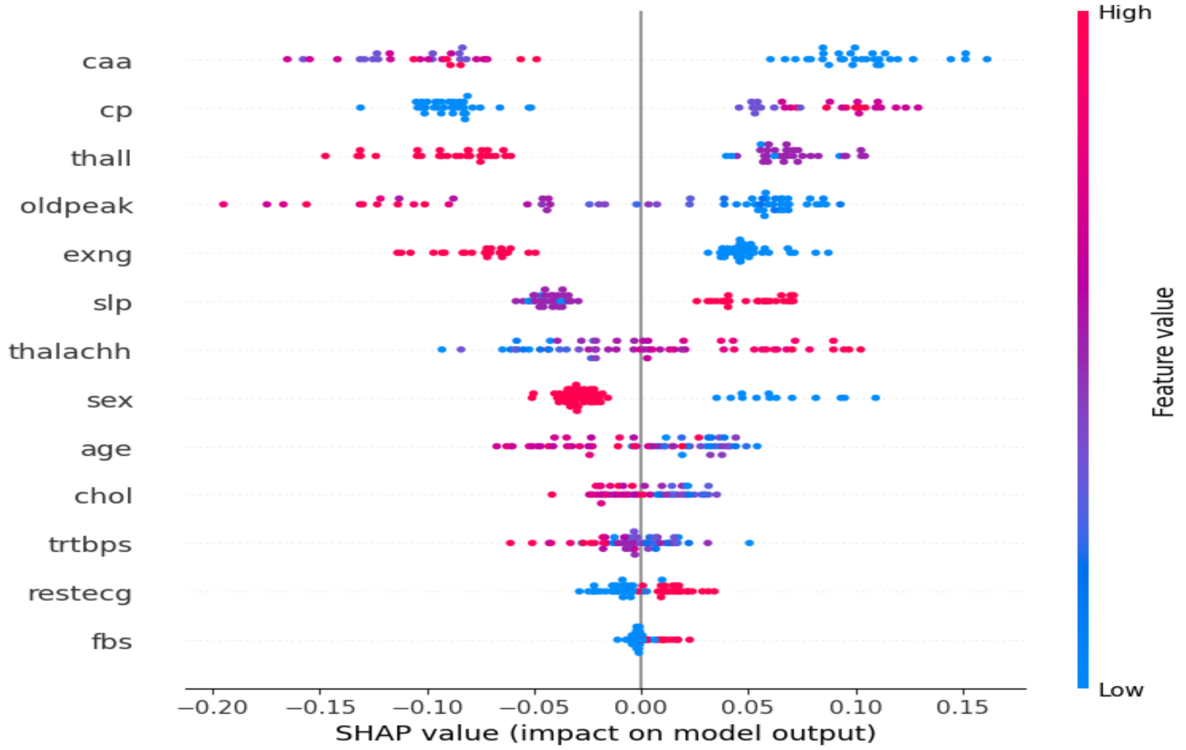
RF	85.24%	84.37%	84.37%	84.37%
KNN	81.96%	80.70%	92%	71.875%

4.2. Local Interpretability Analysis

Summary Plot

The SHAP summary plot visualizes the impact of various features on the output of a machine learning model,i.e., Random Forest model. The attributes on Y-axis represents a feature used in the model, sorted by importance (most important at the top) and attributes on X- axis shows how much a feature influenced predictions. Values to the right (positive) increase the predicted outcome, while values to the left (negative) decrease it.

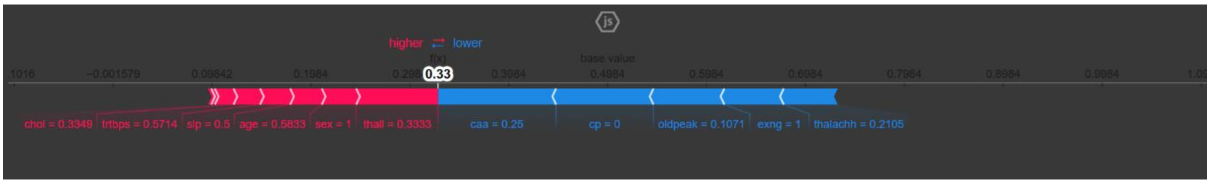
The color represents the actual feature values, blue color shows lower values and red color for higher values. For example,"thalachh" (maximum heart rate achieved) feature effect is higher in value which seems to positively impact predictions.



Force plot

The **force plot** below represents the contribution of different features (variables) to a specific prediction made by Random Forest classifier. The horizontal axis shows the prediction range,i.e., model's average prediction value is 0.4984(base value) whereas the value for final prediction for the instance is 0.33 (output value[f(x)]). Features pushing the prediction higher are represented in red, while those lowering it are in blue.

In this, model's prediction is influenced by features like **thall** and **age** increasing the prediction in negative way, while others like **caa** and **thalachh** are effecting in positive way. The plot helps understand feature-level contributions for individual predictions.



Chapter-5

CONCLUSION AND FUTURE SCOPE OF WORK

5.1. Conclusion

The research on heart disease prediction based on their symptoms by using machine learning classifier is provided in this paper. This model takes the symptom of patients and predict disease by giving output 0 as less chance and 1 as more chance of heart attack. Firstly, I have evaluated performance metrics of all models with and without use of SMOTE method and compared them. The outcome shows that random forest predicts output with higher accuracy, recall and F1 score as compared to other machine learning technique. Secondly, I have selected a model with higher prediction accuracy, i.e., random forest and applied a SHAP explainable approach in local interpretability. The local method used two plots; summary plot that highlighted the most important features in descending order of significance (e.g., "caa," "cp," and "thal") where positive SHAP values indicate features driving predictions higher, while negative values decrease predictions and the force plot that decomposed the model's output, showing the **base value** of 0.33 that shows features on the red side push the prediction higher, while those on the blue side lower it.

5.2. Future work

Expanding the use of SHAP for heart disease detection involves creating models that are not only accurate but also transparent and accessible for clinicians. Researchers could explore combining SHAP with cutting-edge techniques like deep learning to uncover complex patterns in patient data. Additionally, enhancing collaboration between data scientists and healthcare professionals is key to tailoring these models to practical medical needs. Emphasis on real-world validation through clinical trials can help refine these tools, ensuring they deliver reliable and actionable insights in diverse healthcare environments.

Chapter-06

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